Nominal Price Rigidity, Capital Accumulation, and Monetary Policy Transmission

A Reproducible General-Equilibrium Comparison of Flexible and Calvo Prices

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Abstract This paper studies how nominal price rigidity changes the responses of consumption, investment, output, employment, and inflation to technology and monetary-policy innovations in a model with capital accumulation. We compare flexible prices and Calvo pricing within the same monopolistically competitive economy, holding preferences, technology, the steady-state markup, and the monetary rule fixed. Under an illustrative calibration, a positive policy-rule innovation of 0.0025 log units reduces initial output by 0.431 percent and investment by 2.275 percent under Calvo pricing, whereas real allocations are unchanged under flexible prices. Following a 0.01 log-unit technology innovation, output rises by 1.337 percent with flexible prices and 1.250 percent with Calvo prices. Analytic steady-state relations, independent numerical differentiation of equilibrium conditions, and an ordered Schur decomposition corroborate the local first-order rational-expectations solutions. These simulated comparisons are not empirical policy-effect estimates. A second-order local extension also computes illustrative common-capital welfare and physical-capital premia for small shock volatility; no firm-specific-capital equilibrium has been solved.

1 Introduction

In the standard three-equation New Keynesian model, output, inflation, and the nominal interest rate summarize monetary transmission. Investment is absent as a separate choice, however, and the goods-market identity is often $Y_t = C_t$. Incorporating capital accumulation makes it possible to trace the responses of consumption and investment separately. It also disciplines the relationship between current factor prices and expected returns on physical capital. These ingredients matter for interpreting a monetary-policy innovation, even in an otherwise conventional representative-household economy.

Our question is narrowly defined: **How does restricting the frequency of price adjustment change the response of the same economy to the same exogenous shock?** A comparison of perfect competition under flexible prices with monopolistic competition under sticky prices would confound nominal rigidity with a difference in product-market power. Instead, both regimes in this paper share the same monopolistic markup, technology, household preferences, and policy rule. The probability of retaining a previously chosen price is the only structural parameter that changes between the regimes.

We construct a model with capital, one-period nominal bonds, and money in the utility function (MIU). The latter follows the monetary-utility tradition of M. Sidrauski [1]; staggered price setting follows G. A. Calvo [2]. We solve the first-order approximation around a zero-inflation stationary equilibrium, compare impulse response functions (IRFs) to technology and policy-rule shocks, and validate the linear rational-expectations solution using numerical derivatives and an independently implemented ordered Schur procedure. Our comparison complements the standard New Keynesian monetary-transmission framework presented by M. Woodford [3] and J. Galí [4]; it does not claim to introduce a new theoretical mechanism.

The benchmark results are sharp. A positive policy-rule innovation is neutral for real allocations in the flexible-price economy but leads to an initial output decline of 0.431 percent and investment decline of 2.275 percent in the Calvo economy. The same technology innovation initially raises output by 1.337 percent with flexible prices and 1.250 percent with staggered prices. These are **model-generated responses under one illustrative calibration**, not empirical estimates or robust sign restrictions across alternative parameterizations.

2 Economic environment

2.1 Household, assets, and timing

Time is discrete. The representative household chooses consumption C_t, hours N_t, capital K_{t+1}, a position B_t in nominal one-period bonds, and nominal money balances M_t. Preferences are

$$E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\sigma}-1}{1-\sigma} - \chi_n \frac{N_t^{1+\varphi}}{1+\varphi} + \chi_m \frac{m_t^{1-\eta_m}-1}{1-\eta_m} \right], \qquad m_t \equiv \frac{M_t}{P_t}.$$

The consumption and monetary terms are interpreted by continuity at unit curvature. Capital K_t is installed at the beginning of t; the real rental rate is r_t^k and depreciation is δ. Bonds purchased at the end of t at unit nominal price pay $1+i_t$ units of money at $t+1$. Their nominal yield is therefore known at purchase, whereas their real yield is uncertain if next period's inflation is uncertain. The nominal budget constraint is

$$\begin{aligned} P_t C_t + P_t K_{t+1} + B_t + M_t = W_t N_t + P_t(r_t^k + 1 - \delta)K_t \\ + (1+i_{t-1})B_{t-1} + M_{t-1} + D_t + P_t T_t, \end{aligned}$$

where D_t denotes nominal dividends and T_t real net transfers. With interior choices, $\Pi_t = P_t/P_{t-1}$, and

$$\mathcal{M}_{t,t+1} \equiv \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\sigma},$$

the optimality conditions are

$$\begin{aligned} 1 &= E_t\left[\mathcal{M}_{t,t+1}(r_{t+1}^k + 1 - \delta)\right], \\ 1 &= E_t\left[\mathcal{M}_{t,t+1}\frac{1+i_t}{\Pi_{t+1}}\right], \\ \chi_n N_t^{\varphi} &= C_t^{-\sigma} w_t, \\ \chi_m m_t^{-\eta_m} &= C_t^{-\sigma}\frac{i_t}{1+i_t}, \qquad w_t \equiv W_t/P_t. \end{aligned}$$

The second equation prices the inflation-risk exposure of nominal bonds. The fourth equates marginal liquidity services to the nominal opportunity cost of holding money. A non-

Ponzi restriction and suitable transversality conditions complete the household problem; these global conditions are not established by the local linear solution. Because the MIU term is additively separable, real money balances are determined *recursively after* the consumption and nominal-rate allocation in this specification.

2.2 Production, factor markets, and aggregation

Final output is a CES composite of differentiated goods, with elasticity of substitution $\varepsilon > 1$:

$$Y_t = \left(\int_0^1 Y_t(j)^{(\varepsilon-1)/\varepsilon} \, dj \right)^{\varepsilon/(\varepsilon-1)}, \qquad Y_t(j) = \left(\frac{P_t(j)}{P_t} \right)^{-\varepsilon} Y_t.$$

Intermediate producer j employs a Cobb–Douglas production technology. Labor and installed capital services can be reallocated across firms each period. Competitive factor markets imply common real marginal cost mc_t and

$$\begin{aligned} w_t &= mc_t (1-\alpha) \frac{Y_t \Delta_t}{N_t}, \\ r_t^k &= mc_t \alpha \frac{Y_t \Delta_t}{K_t}, \\ Y_t &= \frac{A_t \, K_t^{\alpha} \, N_t^{1-\alpha}}{\Delta_t}, \qquad \Delta_t \equiv \int_0^1 \left(\frac{P_t(j)}{P_t} \right)^{-\varepsilon} dj. \end{aligned}$$

There are no capital adjustment costs and no government purchases. The aggregate resource and capital laws are

$$Y_t = C_t + I_t, \qquad K_{t+1} = (1-\delta) K_t + I_t.$$

The representative household receives intermediate-firm profits, $D_t/P_t = Y_t - w_t N_t - r_t^k K_t$. Government bonds have zero net supply, $B_t = 0$ in equilibrium. The monetary authority supplies money elastically to accommodate its nominal interest-rate rule, and the consolidated public budget requires $P_t T_t = M_t - M_{t-1}$ in this no-public-debt, no-government-spending environment. The paper solves the local real allocation and associated monetary variables; it does not compute a separate exact stochastic path for all nominal public liabilities.

2.3 Flexible prices and Calvo price adjustment

With flexible prices, each monopolistically competitive intermediate producer charges the constant desired markup $\mu = \varepsilon/(\varepsilon - 1)$. Hence

$$mc_t = \mu^{-1} = \frac{\varepsilon - 1}{\varepsilon}, \qquad \Delta_t = 1.$$

Flexible prices do not imply perfect competition here. With Calvo pricing, a firm resets its price with probability $1 - \theta$ and otherwise retains its previous nominal price. Write $p_t^* = P_t^*/P_t$ for the reset price relative to the final-good price and $\lambda_t = C_t^{-\sigma}$. Under common marginal costs, exact reset-price recursions are

$$\begin{aligned} F_t &= \lambda_t mc_t Y_t + \beta\theta E_t\left[\Pi_{t+1}^{\varepsilon} F_{t+1}\right], \\ G_t &= \lambda_t Y_t + \beta\theta E_t\left[\Pi_{t+1}^{\varepsilon-1} G_{t+1}\right], \\ p_t^* &= \frac{\varepsilon}{\varepsilon - 1} \frac{F_t}{G_t}. \end{aligned}$$

The F_t, G_t terms contain marginal utility and therefore do not require another stochastic discount factor on top of the displayed β. Aggregation gives

$$1 = (1-\theta)(p_t^*)^{1-\varepsilon} + \theta \Pi_t^{\varepsilon-1},$$
$$\Delta_t = (1-\theta)(p_t^*)^{-\varepsilon} + \theta \Pi_t^{\varepsilon} \Delta_{t-1}.$$

At a zero-inflation steady state, the first-order Calvo pricing condition is

$$\pi_t = \beta E_t \pi_{t+1} + \kappa_p \widehat{mc}_t, \qquad \kappa_p = \frac{(1-\theta)(1-\beta\theta)}{\theta}.$$

Here $\pi_t = \log \Pi_t$ and $\widehat{mc}_t$ is the log deviation of marginal cost from its stationary value. The slope assumes a *common* marginal cost, not firm-specific capital. Price dispersion vanishes from aggregate production at first order around zero inflation and $\Delta = 1$, but not at second order.

2.4 Monetary policy and disturbances

The central bank follows

$$1 + i_t = (1+\bar{\imath}) \left(\frac{\Pi_t}{\bar{\Pi}} \right)^{\phi_\pi} \left(\frac{Y_t}{\bar{Y}} \right)^{\phi_y} \exp(v_t), \qquad \bar{\Pi} = 1.$$

The rule responds to the deviation of **output from steady state**, not to the gap from flexible-price natural output. Technology and policy-rule shocks evolve according to

$$\log A_{t+1} = \rho_a \log A_t + \varepsilon^a_{t+1}, \qquad v_{t+1} = \rho_v v_t + \varepsilon^v_{t+1}.$$

The shocks are treated as independent. A positive innovation v_t shifts the *rule* upward conditional on endogenous inflation and output; it need not raise the realized equilibrium nominal rate on impact.

3 Steady state and first-order equilibrium

3.1 Analytical stationary allocation

Let $A = 1$, $\Pi = 1$, and $\Delta = 1$. Both pricing regimes share $mc = (\varepsilon - 1)/\varepsilon$. The steady-state relations are

$$r^k = \beta^{-1} - 1 + \delta, \quad \bar{\imath} = \beta^{-1} - 1,$$
$$\frac{K}{Y} = \frac{\alpha mc}{r^k}, \qquad \frac{I}{Y} = \delta \frac{K}{Y},$$
$$\frac{C}{Y} = 1 - \frac{I}{Y}, \qquad \frac{K}{N} = \left(\frac{\alpha mc}{r^k} \right)^{1/(1-\alpha)}.$$

From $Y/N = (K/N)^\alpha$, $w = mc(1-\alpha)Y/N$, and the labor first-order condition,

$$N = \left[\frac{w}{\chi_n} \left(\frac{C}{Y} \frac{Y}{N} \right)^{-\sigma} \right]^{1/(\varphi+\sigma)}.$$

The remaining levels follow from $Y = (Y/N)N$, $C = (C/Y)Y$, $K = (K/N)N$, and $I = \delta K$. Money demand implies $m = [\chi_m C^\sigma (1+\bar{\imath})/\bar{\imath}]^{1/\eta_m}$. The stationary real allocation is identical across the pricing regimes because their desired markup is identical.

3.2 Illustrative calibration

Table Table 1 reports the numerical inputs. These values are deliberately illustrative and have **not** been estimated from microeconomic or macroeconomic data.

Table 1: Parameter values for the benchmark experiment.

Parameter	Value	Parameter	Value
Discount factor β	0.99	Capital share α	0.33
Depreciation δ	0.025	Demand elasticity ε	6
Risk aversion σ	1	Labor curvature φ	1
Labor weight χ_n	1	Money weight χ_m	1
Money curvature η_m	1	Inflation response ϕ_π	1.5
Output response ϕ_y	0.125	Technology persistence ρ_a	0.9
Policy-shock persistence ρ_v	0.8	Calvo non-reset probability θ	0.75
Flexible-price θ	0	Gross steady-state inflation $\bar{\Pi}$	1

The stationary markup is 1.2; $K/Y = 7.8345$, $I/Y = 0.1959$, and $C/Y = 0.8041$. The Calvo Phillips-curve coefficient is $\kappa_p = 0.0858333$.

3.3 Linear equilibrium and determinacy

Write $\hat{x}_t = \log(X_t/\bar{X})$ for positive real quantities and $\tilde{\imath}_t = \log(1+i_t) - \log(1+\bar{\imath})$ for the gross nominal rate. With $s_C = C/Y$, $s_I = I/Y$, and $q = \beta r^k$, the first-order conditions imply

$$
\begin{aligned}
\hat{y}_t &= \hat{a}_t + \alpha \hat{k}_t + (1-\alpha)\hat{n}_t, \\
\widehat{mc}_t &= \sigma \hat{c}_t + (\varphi + \alpha)\hat{n}_t - \hat{a}_t - \alpha \hat{k}_t, \\
\hat{r}^k_t &= \sigma \hat{c}_t + (1+\varphi)\hat{n}_t - \hat{k}_t, \\
\hat{k}_{t+1} &= (1-\delta)\hat{k}_t + \frac{\delta}{s_I}(\hat{y}_t - s_C \hat{c}_t), \\
\hat{c}_t &= E_t \hat{c}_{t+1} - \frac{q}{\sigma} E_t \hat{r}^k_{t+1}, \\
\hat{c}_t &= E_t \hat{c}_{t+1} - \frac{\tilde{\imath}_t - E_t \pi_{t+1}}{\sigma}, \\
\tilde{\imath}_t &= \phi_\pi \pi_t + \phi_y \hat{y}_t + v_t, \\
\eta_m \hat{m}_t &= \sigma \hat{c}_t - \frac{1}{\bar{\imath}} \tilde{\imath}_t .
\end{aligned}
$$

For $\theta > 0$ the Phillips curve is added; at $\theta = 0$ it is **replaced** by $\widehat{mc}_t = 0$ rather than evaluated at a singular slope. After eliminating redundant static variables, the Calvo system has states $(\hat{k}_t, \hat{a}_t, v_t)$ and jump variables $(\hat{c}_t, \pi_t, \hat{n}_t)$. Under flexible prices, the constant-marginal-cost condition eliminates hours and the jump vector is $(\hat{c}_t, \pi_t)$. The system can be written $E_t z_{t+1} = A z_t$.

We check local determinacy using the Blanchard–Kahn root count and the rank of the relevant invariant-subspace projection [5]. An independent ordered QZ decomposition of the *reduced nonsingular transition system* provides a numerical cross-check. This does not amount to an unrestricted solution of a singular generalized linear rational-expectations pencil.

4 Numerical verification and impulse responses

4.1 Independent numerical audit

The analytically derived stationary allocation is substituted into the original nonlinear equilibrium equations. The maximum absolute stationary residual is 1.11×10^{-16}. Independently computed central-difference derivatives agree with the transition matrix used by the simulation code to 5.00×10^{-11} under flexible prices and 9.14×10^{-10} under Calvo pricing. The linear Calvo price condition is separately recovered from the derivatives of the exact reset-price recursions.

Table 2: Independent verification of the reduced equilibrium systems.

Verification statistic	Flexible prices	Calvo pricing
Unstable eigenvalues	2	3
Independent jump variables	2	3
Minimum singular value of rank block	0.5664	0.3264
Maximum transition-matrix difference	5.00×10^{-11}	9.14×10^{-10}
Rank condition	Satisfied	Satisfied

The eigenvalues are 0.8000, 0.9000, 0.9514, 1.0726, and 1.5000 for flexible prices, and 0.8000, 0.9000, 0.9455, 1.0684, 1.6639, and 2.2436 for Calvo prices. Each regime has a locally unique stable solution at the benchmark calibration. The two solution algorithms produce IRFs whose maximum difference across 41 quarters is below 7×10^{-11} log units. All 19 existing audit and model tests passed in the documented benchmark environment. These diagnostics validate the *local first-order solution*, not a globally solved nonlinear stochastic equilibrium.

4.2 Technology innovation

Technology rises unexpectedly by 0.01 log units at date zero and then decays at rate $\rho_a = 0.9$. Output rises by 1.337 percent under flexible prices and 1.250 percent under Calvo pricing in the impact quarter. Investment responds even more strongly: 5.470 percent and 4.948 percent, respectively. At this calibration, limited price adjustment modestly dampens the initial response to a favorable technology innovation.

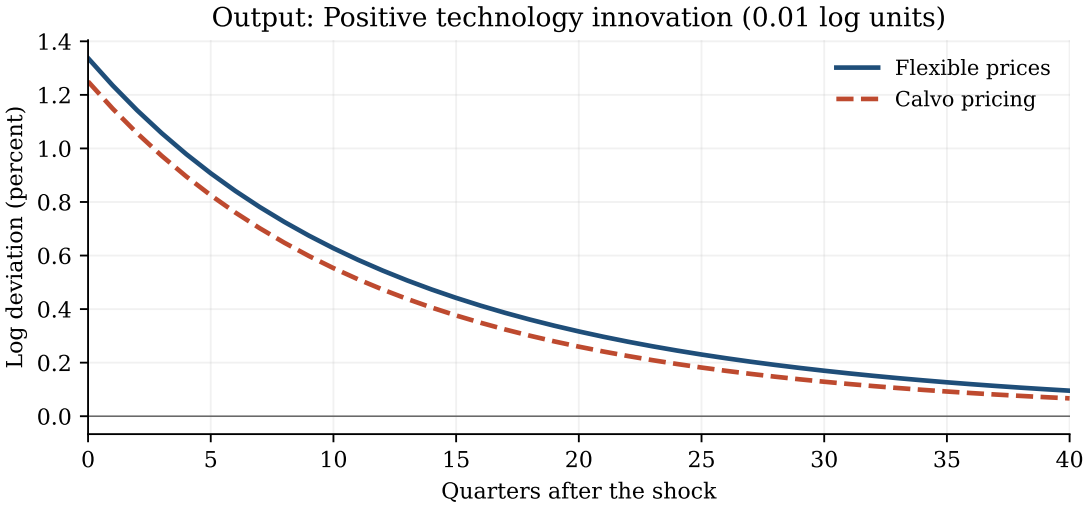


Figure 1: Output response to a positive technology innovation. Both economies share the same technology, desired markup, and policy rule.

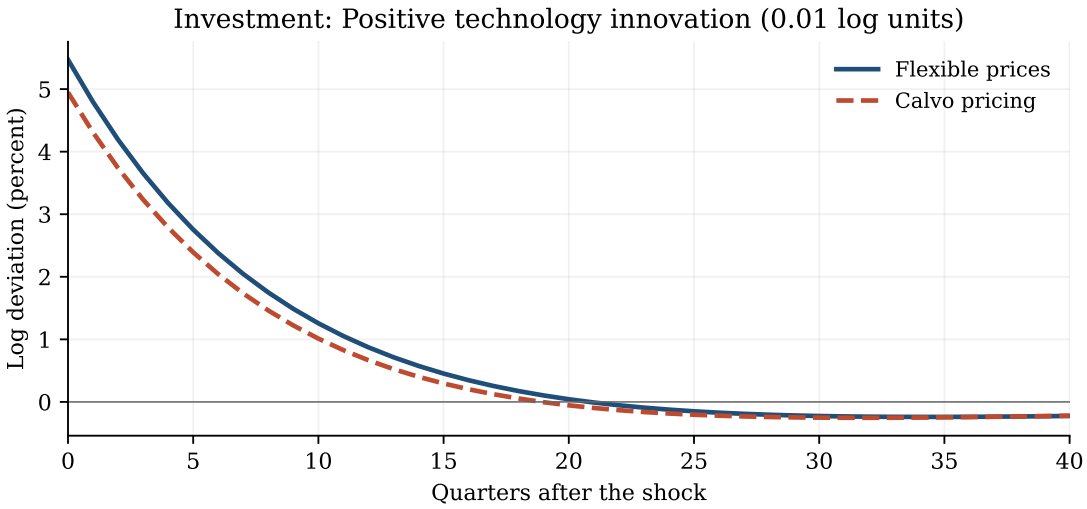


Figure 2: Investment response to a positive technology innovation.

4.3 Monetary policy-rule innovation

The monetary experiment introduces a positive policy-rule residual of $v_0 = 0.0025$ log units of the **gross nominal interest rate**. It is not a fixed increase in the equilibrium nominal interest rate: the feedback rule reacts simultaneously to inflation and output. Real output, investment, consumption, hours, and capital do not respond under flexible prices, consistent with monetary neutrality in this separable MIU model.

With Calvo pricing, output falls by 0.00431472 log units (0.431 percent), investment falls by 0.02274988 (2.275 percent), and hours fall by 0.00643988 (0.644 percent) on impact. Consumption initially **rises** by 0.00017552 log units (0.018 percent), rather than falling. This is a property of the particular frictionless investment technology and calibration; it is not imposed as a sign restriction. The realized nominal interest-rate deviation is negative at impact because endogenous inflation and output responses more than offset the positive innovation to the policy *rule*.

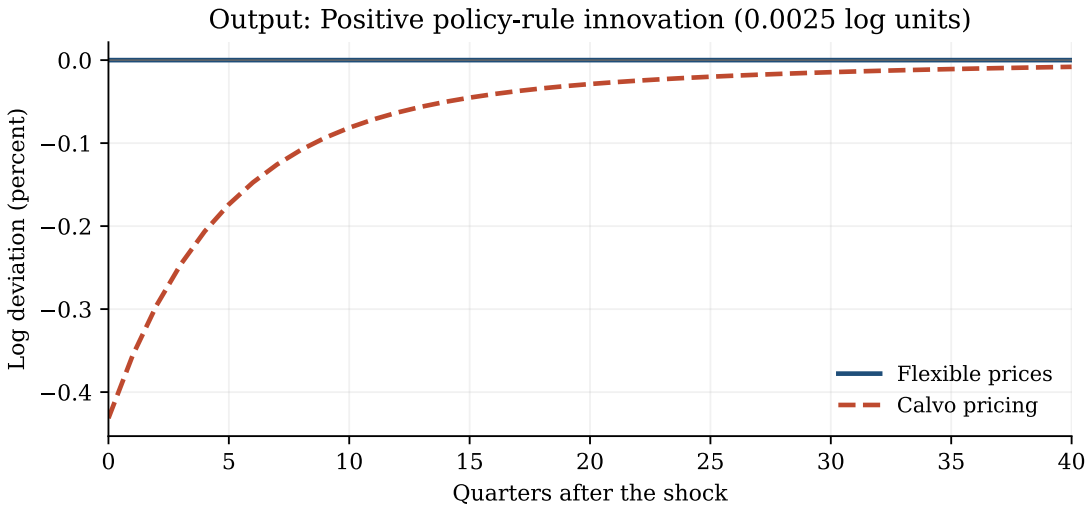


Figure 3: Output response to a positive monetary policy-rule innovation.

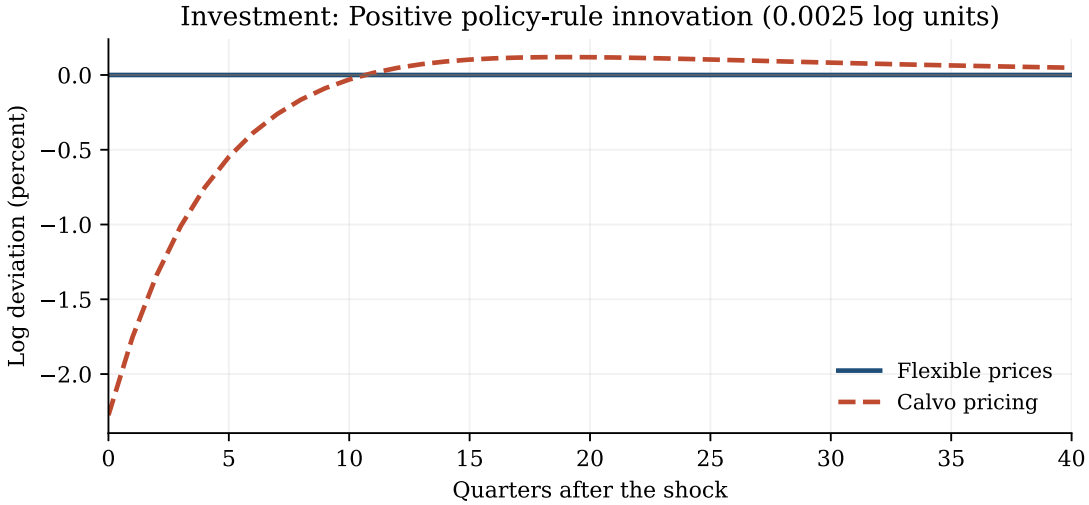


Figure 4: Investment response to a positive monetary policy-rule innovation.

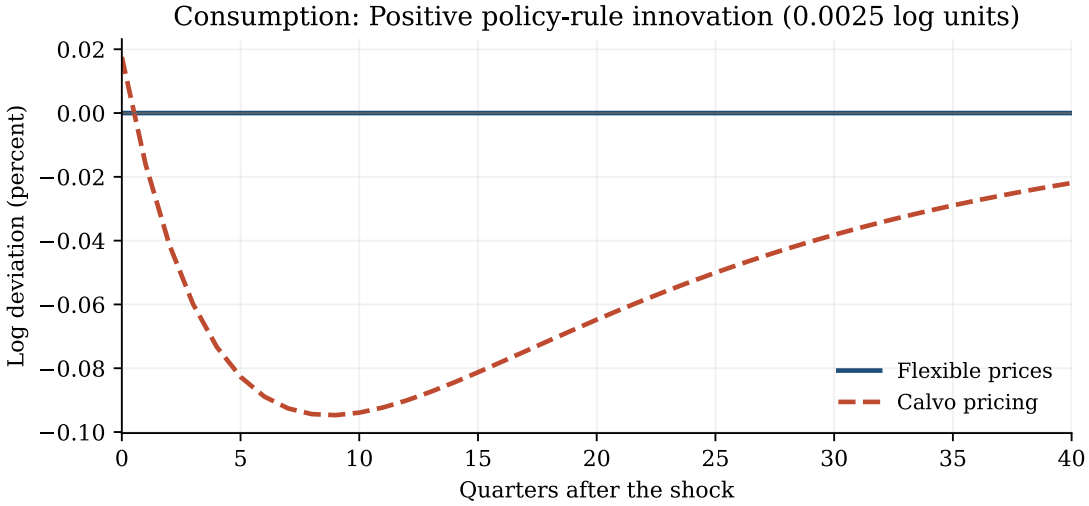


Figure 5: Consumption response to a positive monetary policy-rule innovation.

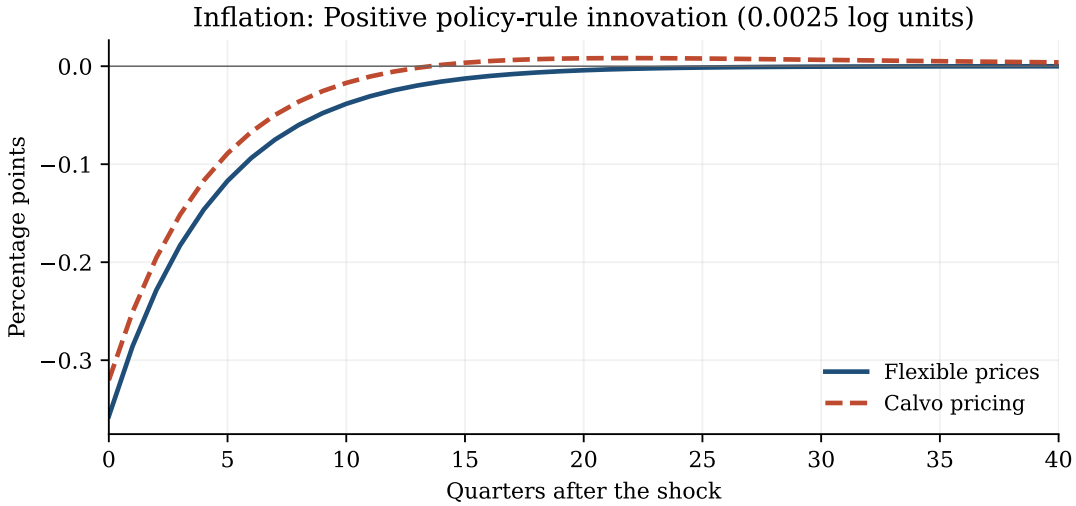


Figure 6: Inflation response to a positive monetary policy-rule innovation. The vertical axis is in percentage points.

Table Table 3 summarizes the initial responses in 100 times log deviations. For quantities this is approximately a percentage change; for inflation and rates the relevant scaling corresponds to percentage points.

Table 3: Impact-quarter responses, in log percent. Technology and monetary innovations are 0.01 and 0.0025 log units, respectively.

Variable	Technology: flexible	Technology: Calvo	Policy: flexible	Policy: Calvo
Output	1.337	1.250	0.000	−0.431
Consumption	0.331	0.349	0.000	0.018
Investment	5.470	4.948	0.000	−2.275
Hours worked	0.503	0.373	0.000	−0.644

The investment response accounts for much of the initial reduction in aggregate demand under nominal rigidity. The comparison isolates **price adjustment frequency**, because monopolistic power is present in both economies. It does not identify a monetary-policy shock in observational data and therefore does not yield an estimated causal policy effect.

5 Second-order stochastic allocation with a common capital market

The first-order model can be extended **without changing the household, factor-market, or monetary-policy specifications**. Here capital services remain freely rentable by all intermediate producers. We retain both nonlinear Calvo pricing recursions and inherited price dispersion and use the benchmark zero-inflation steady state as the point of expansion. This is a computed second-order *local equilibrium*, not an exact global nonlinear solution. Firm-specific installed capital is not introduced in the results of this section.

5.1 Second-order solution and price dispersion

Let $x_t = (\hat{k}_t, \hat{a}_t, v_t, \log \Delta_{t-1})'$ and let $y_t = (\hat{c}_t, \pi_t, \hat{n}_t, \hat{F}_t, \hat{G}_t)'$. Policy coefficients are computed from the nine original equilibrium conditions for x_{t+1}, the capital and nominal-bond Euler

equations, the two Calvo recursion equations, and the contemporaneous Calvo price index. With innovation loading W, the local decision rule is

$$y_t = Hx_t + \frac{1}{2}H_{xx}[x_t, x_t] + \frac{1}{2}H_{\varsigma\varsigma}\varsigma^2 + O(3),$$

and an analogous rule describes the state transition. The ordered generalized Schur decomposition identifies four stable states (one of which is inherited dispersion). Independent comparison with the first-order solver reproduces the original policy coefficients within 10^{-11}. The complete Hessian and the shock-volatility correction satisfy the differentiated system within 10^{-10}. Conditional Gaussian quadrature subsequently checks the *original nonlinear equations*, rather than just the differentiated coefficients.

The dispersion recursion is dynamic:

$$\Delta_t - 1 = \theta(\Delta_{t-1} - 1) + \frac{\varepsilon\theta}{2(1-\theta)}\pi_t^2 + O(3).$$

The coefficient on π_t^2 is 9 for the benchmark parameters. This term enters the second-order resource constraint and is absent at first order only when the initial dispersion is itself of second order. We use pruning to prevent higher-order terms from recursively generating spurious higher-order feedback during simulation.

5.2 Welfare and consumption equivalents

Unlike the textbook quadratic-loss formula around an efficient steady state, our welfare calculation **retains the monopolistic distortion and the utility of real money balances**. For an initial deterministic steady state, conditional expected utility is $W(0) = E_0 \sum_{t\geq 0} \beta^t u(C_t, m_t, N_t)$. Unconditional welfare averages the same value function over the second-order stochastic stationary distribution. We compute both by a quadratic Bellman recursion and obtain the second-order stationary state covariance from a discrete Lyapunov equation. For the benchmark log-consumption utility, a consumption-equivalent change ω solves

$$W^{\text{Calvo}} - W^{\text{flex}} = \frac{\log(1+\omega)}{1-\beta},$$

where only the consumption path of the flexible-price reference is rescaled; hours and liquidity services remain at their reference values.

The second-order local calibration uses **standard deviations** of 0.001 for technology and 0.00025 for the monetary-policy innovation. These are distinct from the *single-shock sizes* in the first-order IRFs. We choose these smaller uncertainty levels because the older, ten-times-larger volatility specification exhibits substantial nonlinear pricing residuals at second order.

Table 4: Second-order common-capital welfare results for the small-volatility illustration. Conditional welfare starts from the shared deterministic steady state; unconditional welfare uses the model-specific stochastic stationary distribution. The negative equivalent is the consumption reduction of the flexible-price reference that makes it as undesirable as the Calvo economy.

Statistic	Flexible prices	Calvo pricing
Conditional expected lifetime utility	548.87266	548.73311
Unconditional expected lifetime utility	548.91212	548.74881
Conditional consumption equivalent, Calvo relative to flexible	—	−0.13945%
Unconditional consumption equivalent, Calvo relative to flexible	—	−0.16317%

5.3 Physical-capital risk pricing

The representative household prices physical capital with the stochastic discount factor $\mathcal{M}_{t,t+1} = \beta(C_{t+1}/C_t)^{-\sigma}$. Denoting by $R^K_{t+1} = r^k_{t+1} + 1 - \delta$ the gross return on the frictionlessly traded capital claim, the real risk-free return is $R^f_t = [E_t\mathcal{M}_{t,t+1}]^{-1}$. A correctly priced equilibrium satisfies both $E_t(\mathcal{M}_{t,t+1}R^K_{t+1}) = 1$ and

$$E_t R^K_{t+1} - R^f_t = -\frac{\mathrm{Cov}_t(\mathcal{M}_{t,t+1}, R^K_{t+1})}{E_t\mathcal{M}_{t,t+1}}.$$

At the deterministic steady state and under the small-volatility calibration, Gaussian quadrature over the second-order decision rules gives the following *local* results:

Table 5: Local capital risk premia and independent Gaussian-quadrature checks. A gross-return premium is not an equity premium; there is no separately priced firm equity claim in these simulations.

Statistic	Flexible prices	Calvo pricing
Expected excess gross capital return (per period)	1.55169×10^{-8}	1.27153×10^{-8}
SDF covariance component (per period)	1.55169×10^{-8}	1.26416×10^{-8}
Absolute capital Euler pricing residual	2.78×10^{-15}	7.29×10^{-11}
Maximum residual across original expected equilibrium conditions	7.86×10^{-15}	4.80×10^{-9}

These small-volatility results pass the Euler-pricing check to a numerical error well below the estimated premium. At the **older** standard deviations (0.01 and 0.0025), however, the Calvo capital Euler residual increases to 7.28×10^{-7}, comparable to the candidate premium, while the largest Calvo recursion residual rises to 4.85×10^{-5}. Thus the corresponding large-

volatility numerical premium is **not** a trustworthy equilibrium calculation from a second-order approximation. The local results above should not be extrapolated to that calibration.

At second order, the leading risk-premium effect is proportional to innovation variance. With constant shock covariance, its **time variation** typically enters at third order; we do not claim a reliable state-dependent conditional premium from the present second-order solution. There is also no equity premium calculation.

5.4 What remains unsolved

A firm-specific-capital economy requires each firm's investment condition and capital shadow value, its optimal Calvo reset price, and the joint distribution of installed capital and relative prices. The common-capital second-order solution does **not** solve these equations, and neither four-economy welfare comparisons nor the associated difference-in-differences have been computed. The numerical evidence above answers only the common-capital sticky-versus-flexible question. The remaining mechanism cannot be approximated merely by multiplying the Phillips-curve slope.

6 Firm-specific capital: a matched-cost first-order comparison

The preceding second-order welfare and risk-premium calculations apply exclusively to the original common-rental-market economy **without investment adjustment costs**. This new exercise constructs a separate *matched-cost* first-order comparison: both capital structures face ((I,K)=_I K(I/K-)^2/2) with the illustrative (_I=4). Hence the results here are **not** directly comparable with the original zero-adjustment-cost IRFs.

We adapt the relative-price and relative-capital conditions from M. Woodford [6], equations (24), (29)–(30), (36)–(44). Woodford assumes sector-specific wages; in the current economy the wage is common across firms. Thus the elasticity of an individual firm's *relative wage* with respect to relative output is zero, while the aggregate wage remains endogenous: (w_t=c_t+n_t). Writing (=/(1-)), (h=) and (_I) for the investment-cost curvature, the local firm rules are

$ k_{t+1}(j)=k_t(j)-p_t(j), p_t^*(j)=-k_t(j), $

where the stable root () is determined by Woodford's quartic, rather than an ad hoc Phillips-curve adjustment. The implied first-order NKPC is

$ *t*=*E*_*t*{t+1}+ *{{}}* _t, +- . $

The aggregate investment Euler equation is modified consistently with the same marginal adjustment cost in both capital structures. At first order the direct resource cost of adjustment vanishes around the symmetric steady state, but its effect on marginal investment decisions does not.

Table 6: First-order matched-adjustment-cost comparison, (_I=4). Technology and monetary innovations are 0.01 and 0.0025 log units, respectively; these are simulated local responses, not estimated causal effects.

Impact effect or structural coefficient	Common capital, Calvo	Firm-specific capital, Calvo
Phillips-curve slope	0.085833	0.022894
Output, monetary-policy innovation (%)	−0.322	−0.832
Investment, monetary-policy innovation (%)	−1.263	−3.236
Inflation, monetary-policy innovation (percentage points)	−0.289	−0.179
Output, technology innovation (%)	1.064	0.764

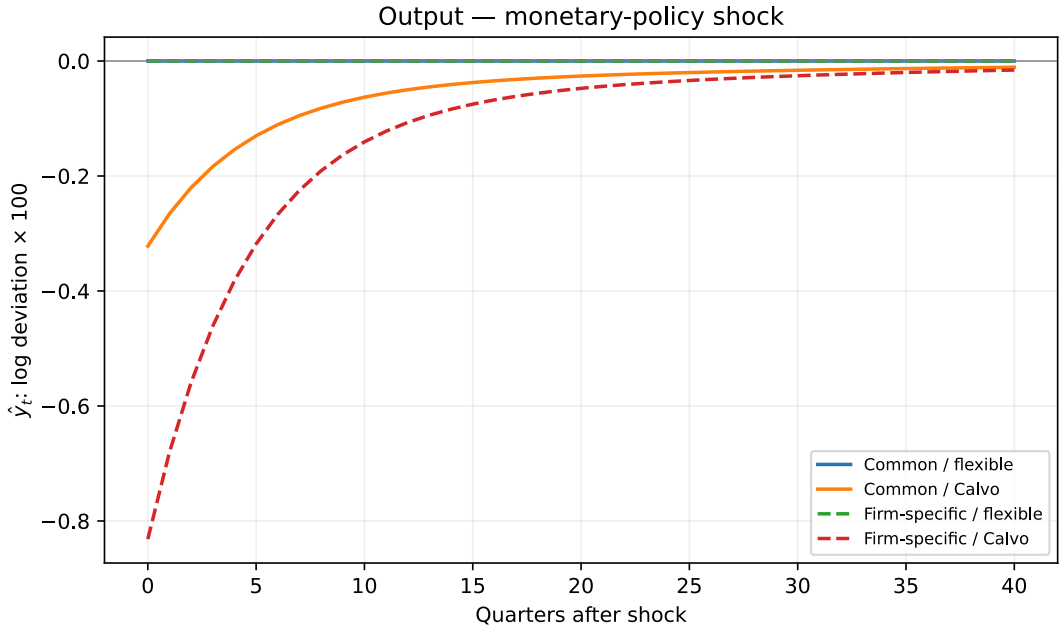


Figure 7: Four first-order output responses to a positive monetary policy-rule innovation. The matched flexible-price paths coincide.

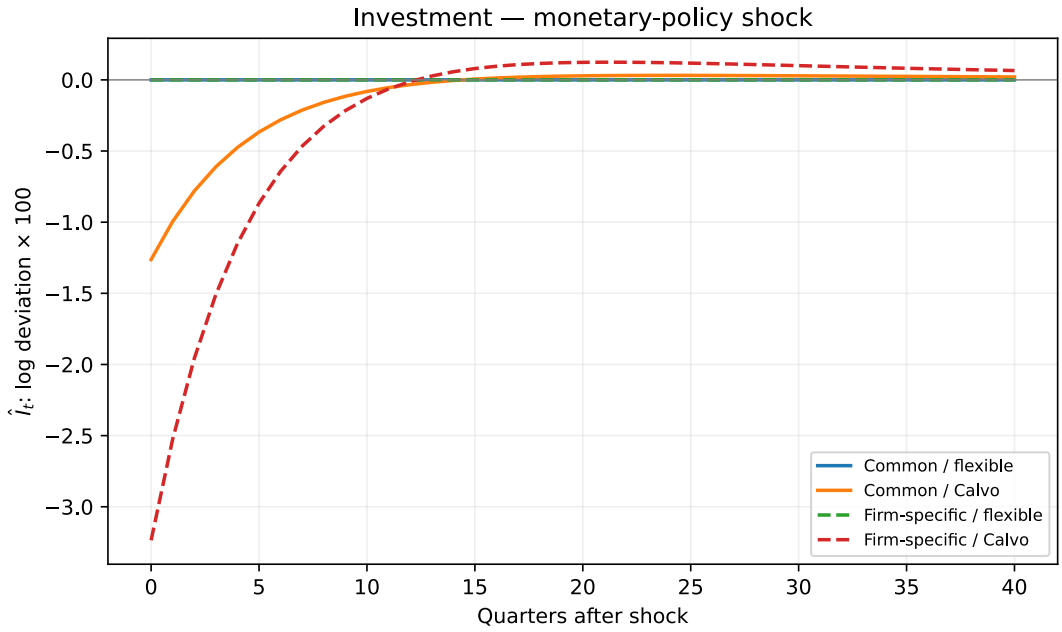


Figure 8: Investment responses to the same monetary-policy-rule innovation under matched investment adjustment costs.

The adapted relative-capital-price system has stable eigenvalues 0.73227 and 0.95155; its algebraic first-order pricing and investment conditions hold to better than (10^{-11}). The four reduced aggregate linear systems satisfy the Schur rank and root-count conditions for the benchmark calibration.

Interpretation and limitations. This is a solved first-order *aggregate reduction* built on the adapted Woodford relative-price equations, **not** a numerical solution of the full nonlinear firm Bellman problem or the joint capital-price distribution. Consequently, neither firm-specific second-order welfare nor comparable capital/equity premia have been computed. The earlier common-capital CEV and premium numbers refer to a different adjustment-cost specification and cannot establish the paper's full welfare-and-asset-pricing research question.

7 Interpretation and limitations

The first-order and local second-order results support the following conclusions. First, keeping the product-market markup fixed avoids a basic confound in sticky-versus-flexible comparisons. Second, capital accumulation makes the investment response visible and quantitatively substantial in the benchmark monetary experiment. Third, independent differentiation, Schur-based solution, rank diagnostics, and Euler-equation residual checks provide corroborating evidence for the local linear numerical solution.

The scope is important. Price dispersion is absent from the first-order resource constraint at the zero-inflation steady state but generates higher-order distortions. A second-order common-capital equilibrium now supplies small-volatility welfare and physical-capital risk-premium calculations, subject to independently reported nonlinear residuals. The older large-volatility approximation does not pass a sufficiently accurate pricing check. The monetary-fiscal closure is specified at the level needed for local equilibrium; a global nonlinear stochastic solution and global household feasibility have not been established. The new second-order estimates still assume commonly rentable capital without investment adjustment costs. Firm-specific capital may change transmission and inflation dynamics but is not solved here. Determinacy is verified for **one calibration** rather than proved

over a parameter region. Finally, global household portfolio feasibility and transversality conditions do not follow from the BK check alone.

The main message is therefore conditional but clear: in a common-capital DSGE economy, introducing Calvo price-setting opportunities generates real responses to a monetary innovation that is neutral under flexible prices. In the specified benchmark, the impact effects are -0.431 percent for output and -2.275 percent for investment. These are reproducible theoretical results, not empirical estimates.

8 Reproducibility

All equations, source code, baseline parameters, independent audit routines, and simulation outputs are available in the project repository, `Hirotaka-Fukui-Pitti-King/dsge-miu-calvo`. The solver in `src/model.py` constructs the steady state and the reduced linear system. `src/audit.py` independently validates its derivatives and Schur solution. `src/second_order.py` and `src/stage2_equilibrium.py` construct the nonlinear Calvo recursions, second-order policy and volatility tensors, welfare recursion, quadrature pricing tests, and pruned stochastic moments. `src/experiment.py` writes the four primary IRF data sets to `results/otkliki_*.csv`; `paper/scripts/make_figures.py` regenerates *English-language SVG figures* from those CSVs. `paper/scripts/check_results.py` tests the numerical values reported in the paper against the CSV output. The workflow runs the model tests, verifies the reported values (including the local second-order results), creates the English figures, and renders this `.qmd` source to a PDF with Quarto and Typst. On the main branch, it commits the compiled PDF into `paper/article.pdf` when the document changes. This pipeline is intended to keep the distributed manuscript tied to the verified model results.

9 Analytical second-order checks and the limits of the current model

We retain these earlier analytic checks as historical diagnostics that motivated the independently solved second-order common-capital model above. The earlier diagnostic covariance computations must not be confused with the newly verified small-volatility equilibrium capital premia. The firm-specific-capital equilibrium and its welfare comparisons remain unsolved. Computing the Hessian of utility alone is not enough: second-order corrections to equilibrium allocations are essential [7].

9.1 Price-dispersion curvature

Holding inherited price dispersion at its zero-inflation stationary value, the exact Calvo price index implies

$$p_t^* = \left(\frac{1 - \theta \Pi_t^{\varepsilon - 1}}{1 - \theta} \right)^{1/(1-\varepsilon)}.$$

Substituting into the exact dispersion recursion and writing $\pi_t = \log \Pi_t$ gives

$$\Delta_t = 1 + \frac{\varepsilon \theta}{2(1-\theta)} \pi_t^2 + O(\pi_t^3), \qquad \Delta_{t-1} = 1.$$

For the benchmark $\varepsilon = 6$, $\theta = 0.75$, the quadratic coefficient is 9. Independent finite-difference calculations verify the zero first derivative and the second derivative of 18. This **local identity** does not replace the full stochastic second-order recursion with inherited dispersion.

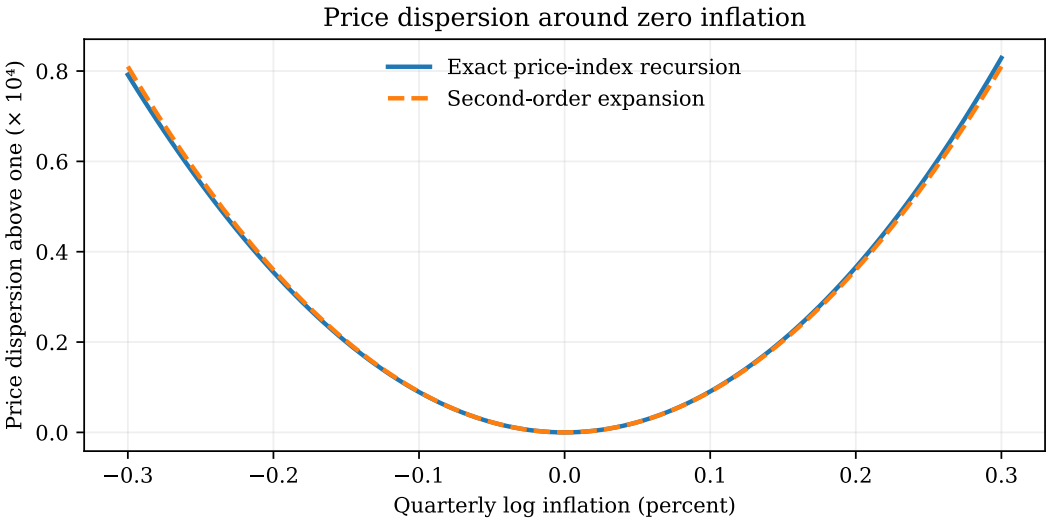


Figure 9: Exact price dispersion and its quadratic expansion, conditional on unit inherited dispersion.

9.2 Firm-specific capital: production-cost identities

Following the distinction in M. Woodford [6], firm-specific installed capital prevents firms from renting capital freely across price vintages. Conditional on fixed installed k_j and output demand y_j, the required employment is

$$n_j = \left(\frac{y_j}{A_t k_j^{\alpha}} \right)^{1/(1-\alpha)}, \quad mc_j = \frac{w_t n_j}{(1-\alpha) y_j}, \quad \left. \frac{\partial \log mc_j}{\partial \log y_j} \right|_{k_j} = \frac{\alpha}{1-\alpha}.$$

Under quadratic investment costs $\Psi(I_j, k_j) = \frac{\phi_I}{2} k_j (I_j/k_j - \delta)^2$, the marginal cost of investment is $\Psi_I = \phi_I (I_j/k_j - \delta)$. The investment first-order condition involves the installed-capital shadow value $q_j = 1 + \Psi_I$. These derivatives have been checked against independent numerical differentiation. They are **not** a solved Calvo firm-level investment policy. A distribution over firm-specific capital and price vintages must be solved before the aggregate Phillips curve, welfare, or impulse responses can be reported for this extension.

9.3 Risk-price covariance diagnostic

Suppose the first-order policy functions imply conditional innovations $d \log C_{t+1} = h_c d\epsilon_{t+1}$ and $d \log r^k_{t+1} = h_r d\epsilon_{t+1}$. With a diagonal innovation covariance matrix Σ_ϵ, the leading covariance component in the capital-premium decomposition is

$$-\frac{\mathrm{Cov}_t(\mathcal{M}_{t,t+1}, R^K_{t+1})}{E_t \mathcal{M}_{t,t+1}} = \sigma \bar{r}^k h_c \Sigma_\epsilon h_r^\top + o(\| \Sigma_\epsilon \|).$$

At illustrative standard deviations of 0.01 for technology and 0.0025 for the monetary-policy-rule shock, this term equals 1.552×10^{-6} in the flexible-price model and 1.264×10^{-6} with Calvo pricing, in gross-return units. Exact Gaussian quadrature applied to the *unadjusted first-order policies* produces nonzero asset Euler residuals of 7.04×10^{-6} and 9.75×10^{-6}, respectively. The pricing errors are larger than the reported covariance components. **These original first-order diagnostic numbers are not equilibrium risk premia**. The separate small-volatility equilibrium calculations presented above use the full second-order policy and explicit asset-pricing residual checks.

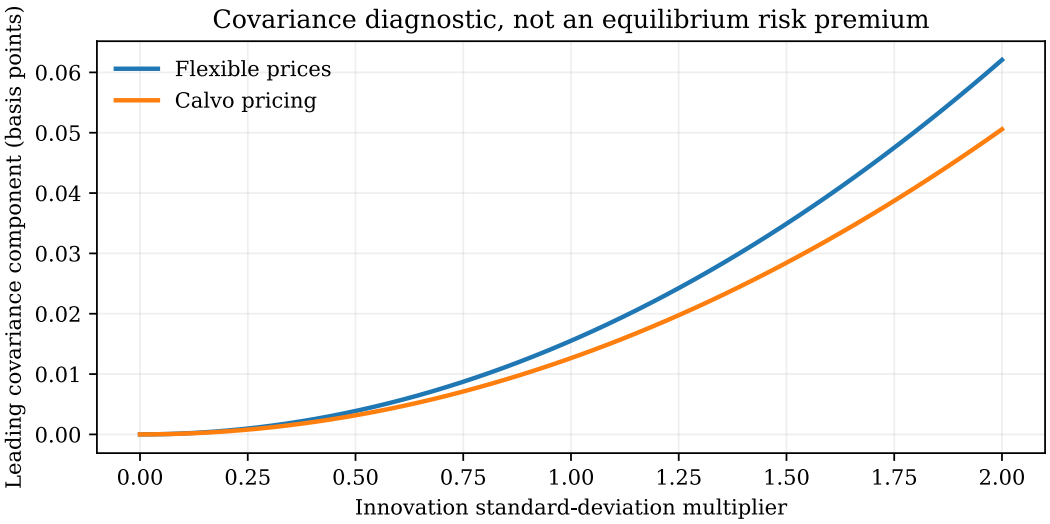


Figure 10: A leading covariance component computed from first-order policy exposures; it is not a priced equilibrium expected excess return.

Consequently, the numerical answer to whether firm-specific capital changes welfare costs and equilibrium premia is **not yet established**, even though the common-capital second-order comparison has been computed. The present checks identify precisely which higher-order pricing and aggregation conditions the next solution must satisfy, rather than substituting diagnostic moments for the sought equilibrium objects.

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